UPI PAYMENTS ANALYTICS & FRAUD DETECTION

End-to-End Data Analytics Project Documentation

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1. INTRODUCTION

Digital payment platforms generate large volumes of transaction data. Analysing this data helps businesses understand payment performance, identify transaction failures, investigate suspicious activities and understand customer behaviour.

The UPI Payments Analytics & Fraud Detection project focuses on analysing 550,000 transaction records using Python, PostgreSQL, SQL and Power BI.

The objective is to transform raw transaction data into meaningful analytical insights that can support business decisions and improve payment operations.

2. PROJECT OBJECTIVES

- Analyse overall UPI transaction volume and value.
- Evaluate successful, failed, reversed and pending transactions.
- Identify payment methods with higher transaction failure rates.
- Analyse transaction patterns across time and geography.
- Investigate fraud patterns using the fraud indicator available in the dataset.
- Perform customer segmentation using RFM analysis.
- Explore customer acquisition and cohort retention patterns.
- Develop interactive Power BI dashboards for business reporting.

3. TECHNOLOGY STACK

Technology	Purpose
Python	Data cleaning and analysis
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Data visualisation
Seaborn	Statistical visualisation
Jupyter Notebook	Exploratory analysis
PostgreSQL	Database management
SQL	Analytical queries
Power BI	Interactive dashboards
Git & GitHub	Version control

4. DATASET OVERVIEW

The project uses a UPI transaction dataset containing 550,000 transaction records.

Metric	Value
Total Transactions	550,000
Unique Customers	79,916
Successful Transactions	506,247
Failed Transactions	21,905
Reversed Transactions	11,019
Pending Transactions	10,829

The dataset was processed to support transaction-level analysis, customer behaviour analysis and fraud pattern exploration.

5. PROJECT METHODOLOGY

The project follows an end-to-end data analytics workflow.

5.1 Data Understanding

- Inspected the dataset structure and column names.
- Analysed data types and distributions.
- Identified missing values and data quality issues.
- Performed initial exploratory analysis.

5.2 Data Cleaning and Preprocessing

- Prepared the transaction dataset.
- Checked data consistency.
- Handled missing values where required.
- Standardised fields for analytical processing.
- Created transaction status indicators.

5.3 Feature Engineering

Created transaction status flags:

- `is_success`
- `is_failed`
- `is_reversed`
- `is_pending`

Generated customer-level features to support behavioural analysis and RFM segmentation.

5.4 Exploratory Data Analysis

Performed analysis of:

- Transaction status distribution.
- Payment method distribution.
- Monthly transaction trends.
- Transaction patterns by time.
- State-wise transaction behaviour.
- Fraud-labelled transaction patterns.

6. PYTHON ANALYSIS

Python was used for data preparation, transformation and exploratory analysis.

Libraries used:

- Pandas

- NumPy
- Matplotlib
- Seaborn

The main analysis was performed using the Jupyter Notebook:

`01_data_understanding.ipynb`

Key outputs included cleaned transaction data, customer features, RFM segments and cohort-related analytical datasets.

7. POSTGRESQL DATABASE IMPLEMENTATION

A PostgreSQL database named:

`upi_payments_analytics`

was created to store the transaction data.

The primary transaction table was:

`upi_transactions`

A total of 550,000 records were successfully loaded and verified.

PostgreSQL was used to perform structured querying and analytical data extraction.

8. SQL ANALYSIS

Seven SQL scripts were developed and executed.

SQL Script	Description
01_payment_performance_analysis.sql	Payment performance analysis
02_advanced_payment_analytics.sql	Advanced payment analytics
03_customer_behavior_analysis.sql	Customer behaviour analysis
04_customer_retention_analysis.sql	Customer retention analysis
05_fraud_detection_analysis.sql	Fraud pattern analysis

06_advanced_sql_window_functions.sql	Advanced window functions
07_export_analytics_datasets.sql	Analytical dataset exports

SQL concepts demonstrated:

- Aggregate functions
- GROUP BY and ORDER BY
- CASE statements
- Common Table Expressions (CTEs)
- Window functions
- Subqueries
- Date-based analysis
- Customer-level aggregations

9. CUSTOMER SEGMENTATION – RFM ANALYSIS

RFM analysis was performed to understand customer behaviour.

The three metrics used were:

Recency: How recently a customer completed a transaction.

Frequency: How frequently a customer performed transactions.

Monetary: The total amount spent by a customer.

Customers were classified into five analytical segments:

1. High Value Customers
2. Loyal Customers
3. Regular Customers
4. At Risk Customers
5. Inactive Customers

This segmentation helps compare customer groups and identify potential opportunities for customer engagement.

10. FRAUD AND RISK ANALYSIS

Fraud analysis was performed using the fraud indicator available in the dataset.

The analysis included:

- Fraud-labelled transaction distribution.
- State-wise fraud analysis.
- Fraud patterns by payment method.
- Fraud analysis by merchant category.
- Comparison of fraud-labelled and non-fraud transactions.

Note: This project analyses dataset-provided fraud labels. It does not implement an independently trained fraud prediction model or independently verify fraudulent transactions.

11. CUSTOMER RETENTION AND COHORT ANALYSIS

Cohort analysis was explored to understand customer acquisition and retention patterns.

The analysis includes:

- Customer grouping by acquisition month.
- Monthly cohort customer counts.
- Cohort retention matrix.
- Month-based retention analysis.

Cohort calculations and retention denominators require validation before the results are used for business decisions.

12. POWER BI DASHBOARD

An interactive Power BI report was developed with five analytical sections.

Page 1 – Executive Overview

- Total transaction volume.
- Total transaction value.
- Payment success rate.
- Failed transactions.
- Fraud-labelled cases.
- Total customers.
- Monthly transaction trends.
- Payment method distribution.

Page 2 – Payment Performance

- Successful versus failed transactions.
- Payment success rate by method.
- Failure rate by payment method.

- Transaction volume by payment method.

Page 3 – Fraud & Risk Analytics

- Fraud cases by state.
- Fraud rate by state.
- Fraud analysis by payment method.
- Fraud patterns by merchant category.
- Fraud versus non-fraud distribution.

Page 4 – Customer Segmentation

- Customer distribution by RFM segment.
- Customer count by segment.
- Average spending by segment.
- Average transaction frequency.
- Customer recency analysis.

Page 5 – Customer Retention & Cohort Analysis

- Customer acquisition by cohort.
- Cohort retention matrix.
- Monthly retention analysis.

13. KEY DASHBOARD METRICS

The Power BI dashboard preview displays the following rounded values:

Metric	Dashboard Value
Total Transactions	550K
Total Transaction Value	16.45bn
Success Rate	92.0%
Failed Transactions	22K
Fraud Cases	5K
Total Customers	79.916K

These are dashboard figures and should be reconciled with the underlying datasets before being treated as audited business results.

14. BUSINESS VALUE

The project demonstrates how transaction data can be used to support business analysis.

Potential business applications include:

- Monitoring payment success and failure rates.
- Identifying payment methods requiring operational investigation.
- Understanding transaction patterns across geographical regions.
- Investigating fraud-labelled transaction patterns.
- Identifying valuable and at-risk customer segments.
- Understanding customer acquisition and retention behaviour.

15. PROJECT LIMITATIONS

- Analysis depends on the quality and availability of the supplied dataset.
- Fraud analysis uses the existing fraud indicator.
- RFM segment classifications are rule-based.
- Cohort retention calculations require further validation.
- Dashboard metrics are rounded for presentation.
- The project does not currently implement a production fraud detection system.

16. FUTURE ENHANCEMENTS

- Develop a machine learning model for fraud prediction.
- Implement customer churn prediction.
- Improve cohort retention calculations and visualisations.
- Introduce automated data refresh.
- Develop additional dashboard filters and drill-through functionality.
- Add automated data quality checks.

17. CONCLUSION

The UPI Payments Analytics & Fraud Detection project demonstrates an end-to-end data analytics workflow using Python, PostgreSQL, SQL and Power BI.

The project covers transaction performance analysis, data cleaning, customer segmentation, fraud pattern exploration and cohort-based analysis.

By combining programming, database management, analytical querying and business intelligence visualisation, the project provides practical experience in transforming raw transaction data into meaningful analytical outputs.

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18. PROJECT LINKS

GitHub Repository:

https://github.com/Kourpreetika/upi-payments-analytics-fraud-detection

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